

0.0 - INTRODUCTION TO ARTIFICIAL INTELLIGENCE

CHAPTER INDEX

- ☐ [AI TOPICS](#)
 - [MACHINE LEARNING](#)
- ☐ [ASSESSING INTELLIGENCE](#)
 - [TURING TEST](#)
 - [BEYOND THE TURING TEST](#)
- ☐ [AI AND GAMES](#)
 - [BRUTE FORCE APPROACHES](#)
 - [MODERN APPROACHES](#)
 - [DEEP LEARNING](#)
- ☐ [A BRIEF HISTORY OF AI](#)
 - [AI TODAY AND CURRENT CHALLENGES](#)
 - [WORRIES ABOUT AI - "SINGULARITY"](#)
 - [SOME OPTIMISTIC INSIGHTS](#)

AI TOPICS

Before diving in the more complex topics, we begin with an introduction to the field of **artificial intelligence**.

First and foremost, let's give a *qualitative* definition of *intelligence*:

INTELLIGENCE

Complex of *psychic* and **mental faculties** that allow man to **think, understand** or **explain facts** or actions, elaborate **abstract models of reality**, **understand and to be understood** by others, to judge, and to render it capable of **adapting to new situations**.

Artificial intelligence refers to the **attempt to imitate human intelligence**.

Thus, it can be defined as follows:

ARTIFICIAL INTELLIGENCE

Partial **reproduction** of the **intellectual property of man** (with particular regard to processes of **learning, recognition, choice**) realized either through the *elaboration of ideal models*, or, concretely, with the *development of machines* that mostly use computers for this purpose.

More rigorously:

📖 AI (Definition by Tom M. Mitchell)

A program is said to learn from **experience** E with reference to some **classes of tasks** T and with **performance measurement** P , if its performance in task $\hat{T} \in T$, as measured by P , improves with experience E .

Overall, *artificial intelligence* is a **broad discipline** that covers several topics, such as:

- Trial and error search,
- Evolutionary computing,
- Knowledge representation and reasoning,
- Automated theorem proving,
- Expert systems,
- Planning, coordination and manipulation,
- Intelligent agents,
- Robotics,
- Natural language processing,
- Vision and speech,
- Machine Learning.

MACHINE LEARNING

We've all heard of *machine learning*. In simple terms, a **machine learning system**:

1. **Learns** from examples during a **training phase**...
2. ... and is (*later*) able to **generalize** and **manage new data** in the **same application domain**.

Machine learning is currently considered one of the most important approaches in artificial intelligence, since **learning is a key component of reasoning**.

Machine learning itself is composed of different *sub-fields*, one of these being **deep learning**.

💡 (TRADITIONAL) MACHINE LEARNING VS DEEP LEARNING

A common *distinction* between *machine learning* and *deep learning* is the following:

- **Machine learning**: humans perform *feature extraction* from *input data* and the system then performs the *classification*.
- **Deep learning**: the system performs both *feature extraction* and *classification*.

ASSESSING INTELLIGENCE

Many key question arise in the domain of artificial intelligence about whether AI is *actually feasible* or, more precisely, *to what extent can it imitate humans*.

🔗 WEAK VS STRONG AI

- **Weak AI:** Is it possible to build machines that act **as if they were smart**?
- **Strong AI:** Is it possible to build machines that **think intelligently**?

TURING TEST

The **Turing test**, originally called "*the imitation game*" by Alan Turing in 1950, is a test of a machine's *ability to exhibit intelligent behaviour equivalent to, or indistinguishable from, that of a human*.

Turing proposed that a human evaluator would judge natural language conversations between a human and a *machine designed to generate human-like responses*.

The evaluator would be aware that one of the two partners in conversation is a machine, and all participants would be separated from one another.

The conversation would be limited to a *text-only channel* such as a computer keyboard and screen so the result would not depend on the machine's ability to render words as speech. If the **evaluator cannot reliably discriminate the machine from the human**, the machine is said to have **passed the test**.

💡 MAIN CRITICISM

Accessing the *immense amount of information available online* can be used to train a machine that answers in an **"only apparently" intelligent way**.

BEYOND THE TURING TEST

To address the above criticism (as well as others), many other tests for AI have been proposed.

An example is given Winograd schemes, for which the solution "requires not only ability of syntactic analysis, but marked **ability of interpretation**".

📖 WINOGRAD SCHEMA

More precisely, a Winograd schema is a *pair of sentences that differ in only one or two words* and that contain an **ambiguity** that is resolved in opposite

ways in the two sentences and *requires the use of world knowledge and reasoning for its resolution*.

The schema takes its name from a well-known example by Terry Winograd:

EXAMPLE

"The city councilmen refused the demonstrators a permit because they feared/advocated violence".

If the word is `feared`, then `they` presumably refers to the city council; if it is `advocated` then `they` presumably refers to the demonstrators.

In his paper, "The Winograd Schema Challenge", Hector Levesque (2011) proposes to assemble a set of such Winograd schemes that are *easily disambiguated* by the human reader (ideally, so easily that the reader does not even notice that there is an ambiguity); *not solvable by simple techniques* such as selectional restrictions; *Google-proof*, that is, there is no obvious statistical test over text corpora that will reliably disambiguate these correctly.

The set would then be presented as a challenge for AI programs, along the lines of the Turing test.

The strengths of the challenge are that it is clear-cut, in that the answer to each schema is a binary choice; vivid, in that it is obvious to non-experts that a program that fails to get the right answers clearly has serious gaps in its understanding; and difficult, in that it is far beyond the current state of the art.

OTHER PROPOSALS

- **Mathematical Games:** they require deep understanding of language, common sense, reasoning skills and multimodal integration.
- **Speech Turing Test:** a benchmark for AI audio systems, measuring whether humans can distinguish between AI-generated and human speech. A system passes this test when it achieves a 50% preference score, meaning listeners can't tell if they're hearing a real person or an AI.
- **Skill acquisition efficiency:** ability of a system to adapt to new problems it has not seen before.

AI AND GAMES

Historically, many algorithms in the field of AI were developed to **solve puzzles** or **win particular games**, as these problems were considered a *good start to develop intelligent systems*.

NOTE

As such, research in AI leveraged knowledge in the field of **game theory**. In general, due to the nature of the domain itself, research in AI has always taken inspiration from many other fields.

BRUTE FORCE APPROACHES

Many **brute force algorithms** have been developed to solve specific problems. Even though brute force approaches have been mostly left behind in modern research, they are still relevant to understand the *evolution of AI solutions*.

A well known example comes from old-school chess game algorithms: **every possible placement** of the pieces on the board is **associated with a score**.

Winning is associated with a $+\infty$ score and defeat with a $-\infty$ score.

When the algorithm has to move, it **evaluates all the possible moves at its disposal**, the opponent's **countermoves** (assuming the optimal player), his moves to the next step, and so on, deeper and deeper **until the end of the game**.

When it is possible to expand all paths to the end of the game, a simple algorithm known as a **minimax** allows the optimal move to always be chosen.

BRUTE FORCE OFTEN ISN'T FEASIBLE

Unfortunately (for non-trivial games) the *tree of all possible alternatives* becomes **too large after a few levels of depth**, therefore the **evaluation** of the score **takes place prematurely**: the choice is **only "locally" excellent**.

The evaluation of the score is complex and characterized by many parameters: how important is a safe position for the king in comparison to a space advantage in the center of the chess board? The optimal values for these parameters were determined by the system itself (learning), analyzing thousands of matches of human champions. The list of openings was provided by chess champions.

MODERN APPROACHES

In 2011, **Watson** (by IBM) won the *Jeopardy* TV quiz.

CONTEXT

Jeopardy! is an American television game show created by Merv Griffin. The show features a quiz competition in which contestants are presented with general knowledge clues in the form of answers, and must phrase their responses in the form of questions.

- Quiz: Napoleon Bonaparte
- Possible answer: Who died in exile in Sant'Elena?

Watson was a calculator (80 TFLOPs in FP32, the same as an NVIDIA rtx 4090), equipped with software for *natural language processing, information retrieval, representation of knowledge, reasoning machine, and machine learning technologies* in the field of the "open domain question answering".

During the quiz, Watson had access to 200 million pages of contents (4 terabytes) all loaded in RAM: encyclopedias (including Wikipedia), dictionaries, thesauri, taxonomies, ontologies (e.g. Wordnet) and newspaper articles.

Watson was **not connected to the internet** (in any case, it would have been too slow to launch online searches).

DEEP LEARNING

Deep learning has been applied numerous times to develop systems capable of beating world class champions in a broad range of games.

Let's take a look at the following timeline:

- **2013 - DeepMind** (Google laboratory) demonstrated the possibility of learning super human skills in old *arcade games* on the Atari console.
- **2016 - AlphaGo** (By Google DeepMind) beats champion Lee Sedol at Go (ancient Chinese game).
- **2017 - OpenAI** developed a bot capable of beating professional gamer Dendy at the game Dota 2.
- **2019 - AlphaStar** (By Google DeepMind) was the first AI to reach the top league of a widely popular e-sport title (*StarCraft 2*).
- **2022 - Meta** built an agent (*Cicero*) that was the first AI to achieve human-level performance in the popular strategy game *Diplomacy*.
- **2025 - NVIDIA** dropped *ACE*, an innovative AI designed to enable autonomous game characters capable of real-time thinking, adaptation and interaction. Its first partnership is the videogame PUBG, where it will perform the same tasks as a real player.

A BRIEF HISTORY OF AI

It is worth taking a look at a brief summary of the evolution of AI.

To do so, let's look at the major events that shaped the field.

1940-1974 - The *birth and golden years*:

- **WW2 era** - The first electronic calculators were built, *Alan Turing* developed his *theory of computation* and *Claude Shannon* introduced *information theory*.
- **1943** - Neurophysiologist *Warren McCulloch* and mathematician *Walter Pitts* modeled the first *simple neural network* using electrical circuits.

- **1949** - Canadian psychologist Donald Hebb introduced the concept of *reinforcement learning*.
- **1951** - Using symbolic reasoning, cognitive scientist *Marvin Minsky* created the first neural network that *solved a problem from the real world*: finding the best way out of a labyrinth.
- **1958** - Psychologist *Frank Rosenblatt* invented the famous *perceptron*, taking inspiration from biological principles.
- **1969** - *Marvin Minsky* and *seymour Papert* published their study "**Perceptrons - An introduction to computational geometry*".

This paper heavily criticised the ability of neural networks built using perceptrons: this study is often thought to have caused a **decline in neural networks research** in the 1970s.

As such, the era **1974 - 1980** is known as **the first winter of AI**: results did not meet expectations and a drastic financing reduction was experienced.

The main problems were related to *poor computational ability* and *small datasets*.

The period **1980 - 1987** saw a **new spring**: it marked the birth of *expert systems* and saw the rebirth of neural networks thanks to the **backpropagation algorithm** (*Rumelhart, Hinton & Williams, 1986*).

"The problem was that, in those times, there was no suitable algorithm to train a reasonably sized neural network. Seeds of the famous *error backpropagation* algorithm existed long before that time. Similar algorithms were used for solving problems in different areas. It was only in the 1970 when this algorithm was applied for training neural networks. Two similar versions were independently developed by Paul Werbos (in his 1974 dissertation) and by David E. Rumelhart, Geoffrey E. Hinton, Ronald J. Williams and James McClelland (1986). The authors of the latter study claimed to have overcome the problems presented by Minsky and Papert, and that "their pessimism about learning in multilayer machines was misplaced". This heralded a Renaissance in the neural networks research. In 1989, Kurt Hornik, Maxwell Stinchcombe and Halbert White published a study to ascertain that multilayer feedforward networks are universal approximators. In other words, however twisted and difficult the class distribution is in $\mathbb{R}^n$, there is a feedforward neural network with a finite structure which is able to approximate the classification regions with any given, fixed precision."

Then came the **second winter (1987 - 1993)**: new funding stopped and specialized hardware was no longer competitive with consumer PCs.

Moving on to the **modern times**, in the period **1993 - 2011** became more and more powerful and advancements were made in *Bayesian Networks* and *intelligent agents*. Thus, classifiers (SVMs - Support Vector Machines) became *more robust*, *multi-classifier* more precise, *Hidden Markov Models* were born and maturity was reached in *feature extraction*.

The consequence was overwhelming success in *vision, biometric systems, speech recognition, robotics, autonomous driving, medical diagnosis, data mining* etc.

AI TODAY AND CURRENT CHALLENGES

As for **nowadays**, *deep learning neural networks* dominate the landscape of AI. They have won numerous challenges and competitions, and have found their use in many applications ranging from *computer vision*, to *bioinformatics and drug testing, medical image analysis* and *natural language processing*.

A lot of these achievements were made possible by a particular architecture of neural networks known as **CNN** (*Convolutional Neural Network*): though it was already developed in 1989 by **Yann LeCun**, in the early stages they were deemed inferior to other techniques due to the *lack of computing power* and *data availability* of the time. As stated above, both of these bottlenecks were no more in modern times, and as a result a CNN called *AlexNet* won the *ImageNet* challenge and *revolutionised computer vision*.

The result was of such significance that Google hired the authors (Geoffrey Hinton and Alex Krizhevsky) and incorporated the technology into its products.

To expand on the importance of CNNs in today's AI research, here are listed a few of the milestones that were made possible:

- **2017 - Mask R-CNN**, a deep neural network for joint object detection and instance segmentation which outputs an entire pixel map and a label map instead of a single class label.
- **2017 - 2018 - Transformers and BERT** Transformers demonstrated that attention can effectively replace recurrence and convolutions in neural networks. BERT showed that pre-training of language models on unlabeled data can be very effective. However, it is still easy to make these systems fail and make them not pass the Turing test.
- **2018 - Turing Award** was awarded to the *founding fathers of deep learning* Yoshua Bengio, Geoffrey Hinton and Yann LeCun.
- **2016-2020 - 3D Deep Learning** From 2016 on, the first models capable of successfully outputting 3D representations were developed. Furthermore, prediction of 3D models became possible from a single 2D image.
- **2020 - GPT-3** This year, GPT-3, the first version of OpenAI's language model, was released. It upscaled existing models to 175 billion parameters and came with a *text-in / text-out* interface to ease use cases in coding, poetry, blogging and news articles.

🧐 WHAT'S NEXT?

There are still **numerous challenges** to tackle to reach the next generation of AI, such as *un- or self-supervised learning, interactive learning, accuracy, robustness and generalisation, inductive biases*, understanding and

mathematics, memory and compute, and last but not least, ethics and legal questions.

It also remains open whether AI will keep growing at this rate in the number of parameters or if they will reach a plateau.

WORRIES ABOUT AI - "SINGULARITY"

Ethics and morality shall not be underestimated in relation with AI research and development.

Here, we present a few examples that raise concerns and questions about our use of this technology, especially regarding biases in the training data.

Norman - The Psychopath AI

We present you **Norman**, world's first **psychopath AI**.

Norman is born from the fact that the *data that is used to teach* a machine learning algorithm can *significantly influence its behavior*.

So when people talk about AI algorithms being **biased and unfair**, the **culprit is often** not the algorithm itself, but **the biased data** that was fed to it. The same method can see very different things in an image, even sick things, if trained on the wrong (or, the right!) data set.

Norman suffered from **extended exposure to the darkest corners of Reddit**, and represents a case study on the dangers of Artificial Intelligence gone wrong when **biased data** is used in machine learning algorithms.

Gab's Chatbot

A **far-right social media company** (named Gab) created AI chatbots that **deny the Holocaust**.

The broader array of Gab's AI bots are easily goaded into **parroting extremist antisemitic and white supremacist beliefs**, as well as conspiratorial disinformation, including that Covid-19 vaccines contain "*nanotechnology that could potentially be used to track and control human behaviour*."

As a platform, Gab has a *dark history with antisemitism*. The social media site first burst into national consciousness in 2018, after it was revealed the shooter in the Tree of Life synagogue massacre in Pittsburgh had been posting hateful screeds to Gab, including reportedly posting this message shortly before killing 11 worshippers.

AI generated songs

In 2024 a **song about immigrants** whose music, vocals and artwork were entirely **generated using artificial intelligence** has made the Top 50 most listened to songs in Germany, in what may be a first for a leading music market.

Verknallt in einen Talahon is a *parody song* that weaves modern lyrics, many of them based around racial stereotypes about immigrants, with 60s schlager pop.

Other worries are centred around what's referred to as *the Singularity*.

SINGULARITY

A technological singularity is a point, predicted in the development of a civilisation, where **technological progress accelerates beyond the ability to understand** and predict of human beings: - from the moment that such calculator will be built: intelligent human beings will witness an **exponential development**, enabling the design of new systems to be delegated to the systems themselves (which operate, with positive feedback, effortlessly 24 hours a day!).

The worry comes from the prediction that AI will keep scaling up at impressive rates and researchers eventually won't be able to understand and predict its behaviour.

One of the most difficult questions in the future of AI will involve precisely the possibility that our machines can know things that we ourselves cannot comprehend.

-Nello Cristianini

SOME OPTIMISTIC INSIGHTS

It's also important to emphasise how much AI has contributed to advancements in countless research fields.

Examples of such contributions are the 2024 Nobel prize in physics and 2024 Nobel Prize in chemistry, both related to neural networks.

- **Physics** - John J. Hopfield and Geoffrey E. Hinton were awarded the Nobel Prize in Physics for their foundational contributions to machine learning, such as the Hopfield network.
- **Chemistry** - Demis Hassabis and John Jumper won the 2024 Nobel Prize in Chemistry for protein structure prediction with AlphaFold, and to David Baker for computational protein design. This is a monumental achievement for AI. Google DeepMind open-sources AlphaFold 3, ushering in a new era for drug discovery and molecular biology.